

A Self-Closing Gate Conundrum

#1 Victory Royale

Element H: Prototype Testing and Data Collection Plan

Student G's family needs a self-closing mechanism that closes their family's gate on its own because his family does not want unwanted animals and people to get inside their fence gate because it is a safety concern that could cause injury. This happens because student G's gate can not close on its own.

The tests that we will run on our prototype will give us the information we need to make informed decisions about iterating our prototype to make it work better in the future. For the closing power of the spring we will first attach the spring to the gate and the fence. For the affordability test we will add up the cost of our materials that we have bought and compare the total cost to our design requirements.

Testing procedures

Weight

1. Weigh with digital scale.
2. Repeat step one, four times.
3. Record data.

Measurements

1. Measure with a yardstick.
2. Repeat step one, four times.
3. Record data.

Gate closing power

1. Apply spring and clamp to the gate.
2. Open gate normally.
3. Open gate forcefully.
4. Repeat step two and three four times.
5. Record data about how the spring and clamp react to each type of gate opening.

Cost

1. Find all material costs.
2. Add material costs.
3. Repeat step one and two, four times.
4. Record data.

The tests that we will run on our prototype will give us enough information to design and create a new iteration of the previous prototype because of the information that we have gathered by doing our tests multiple times to get a better understanding of how we can improve our design.

The feedback from Ms. Z was that we needed to alter our design requirements for clarity, and that we needed to double check our testing procedures so that they will meet at least 60% of our design requirements. Ms. Z is qualified to look over our testing plan because she is our science and engineering teacher, so she has personal experience doing testing plans.